

Moreom Akter

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ORCID | Google Scholar | LinkedIn | Personal Website

Research Interests

• Galaxy Evolution • AGN Feedback • Multi-wavelength Astronomy • Dark Matter • Large-Scale Cosmology • Computational Astrophysics

Education

Master of Science in Physics

Astrophysics Concentration, Thesis-based

GPA: 3.824/4.00

Thesis: The Influence of Active Galactic Nuclei (AGN) Feedback on Star Formation in Spiral Galaxies.

Thesis Advisor: Dr. Wayne A. Barkhouse

University of North Dakota

August 2022 - May 2025

Master of Science in Physics

Astrophysics Concentration

GPA: 3.612/4.00

Advisor: Dr. Esteban D. Araya

Western Illinois University

January 2021 - July 2022

Bachelor of Science in Physics

Physics

GPA: 3.18/4.00

4th position out of 25 students; top 16% of cohort

Kabi Nazrul Government College, Dhaka, Bangladesh

January 2013 - September 2018

Research Experience

Western Illinois University

Graduate Research Assistant

June 2021 - May 2022

Macomb, IL, USA

- Research project: *Characterization of Ionized Jet Candidates in High-Mass Star-Forming Regions Using VLASS.*
- Analyzed 3 GHz VLASS radio continuum data for 90 ionized-jet candidates; processed FITS images using CASA and developed Python/Jupyter workflows for flux-density, RMS, and spectral energy distribution (SED) analysis.
- Compared multi-epoch VLASS observations and incorporated THOR 1–2 GHz data to investigate radio-continuum variability and spectral indices, distinguishing thermal and non-thermal emission.
- Presented the research at the 240th AAS Meeting (2022) and the Illinois State Academy of Science Annual Meeting, where the oral presentation received 2nd place.

Publications

- **Akter, M.**, Barkhouse, W. A., Kalawila Vithanage, S. P., Gamage, G. L., López-Cruz, O. “The influence of AGN feedback on star formation in red spiral galaxies.” *New Astronomy*, 125, 102535 (2026). doi:10.1016/j.newast.2026.102535.
- Lead author.
- Conducted multi-wavelength analysis using SDSS, Chandra, and XMM-Newton data to investigate AGN feedback and galaxy evolution.
- Barkhouse, W. A., Kashur, L., **Akter, M.**, Kalawila Vithanage, S. P., Gamage, G. L., López-Cruz, O. “Properties of Cluster Red-Sequence Spiral Galaxies.” *New Astronomy*, 127, 102588 (2026). doi:10.1016/j.newast.2026.102588.

- Co-author.
- Contributed to multi-wavelength photometric and spectroscopic analysis of cluster spiral galaxy populations using GALEX, WISE, and CFHT data.

Conference Presentations

Poster Presentations

- **Akter, M.** “The Influence of AGN Feedback on Star Formation in Spiral Galaxies.” ND EPSCoR Annual Conference, Alerus Center, Grand Forks, ND, November 21, 2024. Abstract available.
- **Akter, M.,** Araya, E. D., Alkishnawi, F., Hofner, P. “Characterization of Ionized Jet Candidates in High-Mass Star-Forming Regions Using VLASS.” iPoster, 240th Meeting of the American Astronomical Society (AAS), Pasadena, CA, June 12–16, 2022. Abstract published in *Bulletin of the American Astronomical Society*, Vol. 54, No. 6 (2022). ADS: 2022AAS...24010517A.

Oral Presentations

- **Akter, M.** “Investigating the Spectral Energy Distribution of Young High-Mass Stellar Objects Using VLASS.” 113th Illinois State Academy of Science Annual Meeting, Macomb, IL, April 9, 2022. Awarded **2nd Place** in the student competition. Award recipient List.

Teaching Experience

University of North Dakota

Graduate Teaching Assistant

August 2022 - May 2025

Grand Forks, ND, USA

- Lab instructor and grader for College Physics A and B Laboratories.
- Lab instructor and grader for Introductory Astronomy Laboratory across multiple sections.
- Mentored undergraduate students in experimental techniques, scientific reporting, and quantitative analysis.

Western Illinois University

Graduate Teaching Assistant

January 2021 - May 2021

Macomb, IL, USA

- Provided tutoring support for undergraduate physics courses.

Peer Review and Professional Service

NASA’s NICER Cycle 7 Proposal Review Panel

Executive Secretary

November 12–15, 2024

National NASA Peer Review Panel

- Served as Executive Secretary on one of five national review panels for NASA’s Neutron Star Interior Composition Explorer (NICER) mission proposals.
- Supported the panel chair and coordinated discussions among panel members.
- Facilitated communication with NASA representatives.
- Compiled ranked lists and prepared conclusion notes for proposals.
- Participated in training for efficient and fair federal peer-review processes.

Technical Skills

Programming	Python, R
Scientific Computing	Python/Jupyter, Mathematica, Matplotlib
Astronomical Software	CASA, SAOImageDS9, Glue, Astrometrica, Stellarium
Data Analysis	GAWK, Gnuplot, spectral-index analysis, SED modelling
Astronomical Data	SDSS, Chandra, XMM-Newton, GALEX, WISE, CFHT, VLASS
Machine Learning	Learning machine learning through Coursera.
Typesetting and Office	L ^A T _E X, Microsoft Office, Microsoft Excel

Awards and Scholarships

- **Angle Fund Scholarship**, University of North Dakota, 2024.
- **Open Door Scholarship**, University of North Dakota, 2023.
- **Chairman’s Award for Excellence in Research**, Western Illinois University, 2022.
- **Physics Graduate Scholarship**, Western Illinois University, 2021.
- **2nd Place, Student Competition**, 113th Illinois State Academy of Science Annual Meeting, 2022.
- **Bronze Medal**, International Astronomy and Astrophysics Competition, 2020.

Professional Memberships and Leadership

Treasurer, Bangladesh Student Association 2024–2025	<i>University of North Dakota</i> <i>Grand Forks, ND, USA</i>
Managed organizational finances and led fundraising activities, raising approximately \$1,000. Organized cultural programs and large student events, including BDSA Night.	
Secretary, Bangladesh Student Association 2023–2024	<i>University of North Dakota</i> <i>Grand Forks, ND, USA</i>
Supported organizational activities, student engagement, and cultural programming.	
Volunteer Member, UND International Organization 2023–2025	<i>University of North Dakota</i> <i>Grand Forks, ND, USA</i>
Volunteered at the 62nd Annual Feast of Nations, supporting international cultural outreach.	
• Student Member, American Astronomical Society, 2021–2025. • Early Career Member, American Physical Society, 2020–Present. • National Volunteer, International Astronomical Union, 2020–2021.	

STEM Outreach and Public Engagement

STEM Outreach Instructor – BarinSTEM Event October 10, 2024	<i>University of North Dakota</i> <i>Grand Forks, ND, USA</i>
• Led hands-on physics demonstrations on electromagnetic induction using the jumping-ring apparatus. • Conducted Tesla coil demonstrations with noble gases (He, Ne, Ar, Xe) to illustrate plasma phenomena. • Outreach activity featured in a University of North Dakota press release.	
STEM Outreach Leader –University STEM Camp July 2021	<i>Western Illinois University</i> <i>Macomb, IL, USA</i>
• Delivered an outreach session titled “Exploring the Hidden Universe” for high school students.	

- Guided students in citizen-science research using Zooniverse and LOFAR Galaxy Zoo to identify supermassive black holes and star-forming galaxies.
- Demonstrated basic spectroscopy using diffraction gratings and light pens.
- Developed educational materials and an outreach website using Google Sites.

Departmental Outreach Presenter

Western Illinois University

March 2021

Macomb, IL, USA

- Presented an outreach talk titled “Exploring the Night Sky: Past, Present, and Future.”
- Demonstrated planetary alignment using Stellarium software.
- Explained techniques for observing the International Space Station (ISS).
- Created an educational outreach website using Google Sites.

Additional Astronomy Activities

- **International Astronomy and Astrophysics Competition**, 2020 – Bronze Medal.
- **International Astronomical Search Collaboration**, 2020 – Identified one asteroid using Astrometrica.

Languages

English	Fluent; academic and research proficiency
Bangla	Native
Hindi	Speaking fluently

References

Available upon request.